

Let $A, B \subseteq \Sigma^*$

Statement: if A and B are regular then $A \cap B$ is regular.

$$\exists M_1 \text{ s.t. } L(M_1) = A ; \exists M_2 \text{ s.t. } L(M_2) = B$$

To construct M_3 s.t. $L(M_3) = A \cap B$.

M_3 runs over strings in Σ^* ; so $L(M_3) \subseteq \Sigma^*$

$$M_1 = (Q_1, \Sigma, \delta_1, \delta_1, F_1) \quad M_2 = (Q_2, \Sigma, \delta_2, \delta_2, F_2)$$
$$M_3 = (Q_3, \Sigma, \delta_3, \delta_3, F_3).$$

$$Q_3 = Q_1 \times Q_2 = \{ (q_1, q_2) \mid q_1 \in Q_1 \text{ and } q_2 \in Q_2 \}.$$

$$\delta_3 = (\delta_1, \delta_2)$$

$$F_3 = F_1 \times F_2 = \{ (q_1, q_2) \mid q_1 \in F_1 \text{ and } q_2 \in F_2 \}.$$

$$\delta_3: Q_3 \times \Sigma \rightarrow Q_3$$

$$\delta_3 \left(\underbrace{(q_1, q_2)}_{\in Q_3}, a \right) = \underbrace{\left(\overbrace{\delta_1(q_1, a)}^{\in Q_1}, \overbrace{\delta_2(q_2, a)}^{\in Q_2} \right)}_{\in Q_3}$$

M_3 - product of M_1 & M_2 .

To prove $L(M_3) = A \cap B$

Theorem. $L(M_3) = L(M_1) \cap L(M_2)$

Lemma 1. For all $x \in \Sigma^*$,

$$\hat{S}_3(\underbrace{(q_1, q_2)}_{\in Q_3}, x) = (\underbrace{\hat{S}_1(q_1, x), \hat{S}_2(q_2, x)}_{\in Q_3})$$

$\in Q_1 \qquad \in Q_2$

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$\underbrace{\hspace{10em}}_{\in Q_3}$

Proof. Induction on $|x|$.

Base case $x = \epsilon$; $\hat{S}_3((q_1, q_2), \epsilon) = (q_1, q_2) = (\hat{S}_1(q_1, \epsilon), \hat{S}_2(q_2, \epsilon))$

Induction step.

$$\begin{aligned} \hat{S}_3((q_1, q_2), xa) &= (\hat{S}_1(q_1, xa), \hat{S}_2(q_2, xa)) \\ &\quad \text{To prove} \\ &= \hat{S}_3(\hat{S}_3((q_1, q_2), x), a) \quad \text{Definition of } \hat{S}_3 \\ &= \hat{S}_3((\hat{S}_1(q_1, x), \hat{S}_2(q_2, x)), a) \quad \text{Induction hypothesis} \\ &= (\hat{S}_1(\hat{S}_1(q_1, x), a), \hat{S}_2(\hat{S}_2(q_2, x), a)) \\ &\quad \hookrightarrow \text{Definition of } \hat{S}_3. \\ &= (\hat{S}_1(q_1, xa), \hat{S}_2(q_2, xa)) \quad \text{Definition of } \hat{S}_1 \text{ and } \hat{S}_2. \end{aligned}$$

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Proof. For all $x \in \Sigma^*$

$$x \in L(M_3) \Leftrightarrow \hat{\delta}_3(\delta_3, x) \in F_3 \quad [\text{acceptance}]$$

$$\downarrow \Leftrightarrow \hat{\delta}_3((\delta_1, \delta_2), x) \in F_1 \times F_2 \quad [\text{Def of } \delta_3 \notin F_3]$$

$$\Leftrightarrow (\hat{\delta}_1(\delta_1, x), \hat{\delta}_2(\delta_2, x)) \in F_1 \times F_2$$

$$\Leftrightarrow \hat{\delta}_1(\delta_1, x) \in F_1 \text{ and } \hat{\delta}_2(\delta_2, x) \in F_2 \quad [\text{Lemma 1}]$$

[definition of set product]

$$\Leftrightarrow x \in L(M_1) \text{ and } x \in L(M_2). \quad [\text{Acceptance}]$$

$$\Leftrightarrow x \in L(M_1) \cap L(M_2) \quad [\text{Defn. of intersection}]$$

$$x \in L(M_3) \text{ iff } x \in L(M_1) \cap L(M_2).$$

Question. Are regular sets closed under
Complementation?

Let $A \subseteq \Sigma^*$. if A is regular is $\bar{A}$ regular? Yes

$$\exists M \text{ s.t. } L(M) = A \quad \exists \bar{M} \text{ s.t. } L(\bar{M}) = \bar{A}$$

Interchange the set of accept states and non-accept states.

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Yes.

$$A \cup B = \overline{(\bar{A} \cap \bar{B})}$$

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An explicit construction

From $M_1 [L(M_1) = A]$ and $M_2 [L(M_2) = B]$ construct M_3 s.t. $L(M_3) = L(M_1) \cup L(M_2)$.

Intersection $F_3 = F_1 \times F_2$

Union $F_3 = (F_1 \times Q_2) \cup (Q_1 \times F_2)$

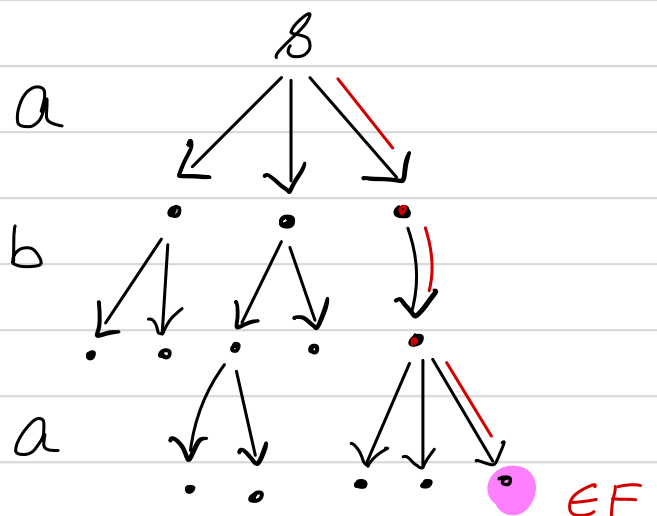
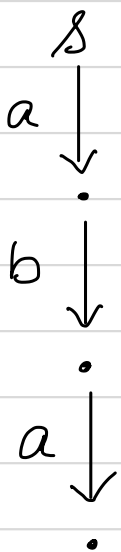
$$F_3 = \{ (q_1, q_2) \mid q_1 \in F_1 \text{ or } q_2 \in F_2 \}.$$

Non-determinism

- Next State of a Computation is not uniquely determined by current state.
- Important Concept in the design of efficient algorithms.
- There are many combinatorial problems with efficient non-deterministic solution but no known efficient deterministic solution.

$$P \stackrel{?}{=} NP$$

DFA-M - $S: Q \times \Sigma \rightarrow Q$.	NFA-N - S is non-deterministic $S: Q \times \Sigma \rightarrow 2^Q$.
$x = a b a$ Run of M on x	Run of N on x



N accepts x if at least one computation path starting from at least one initial state leads to a final state.

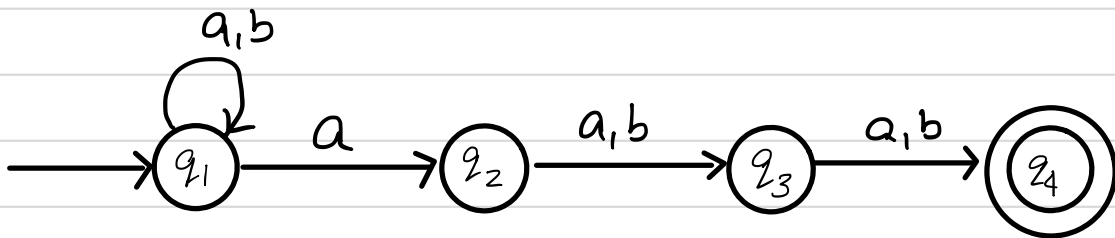
Example

$A = \{x \in \{a,b\}^* \mid \text{the third symbol from right is } a\}$

$abababb \in A$

$ababbab \notin A.$

NFA N :



Claim: $L(N) = A.$